Fiscal Year:	FY 2011	**Task Last Updated:**	FY 10/28/2010
PI Name:	Olson, Sandra Ph.D.		
Project Title:	Correlation of Normal Gravity Aerospace Materials Flammability Data with Flammability in Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Zip Code:	44135	**Congressional District:**	9
Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hirsch, David (NTEC, Jacobs Engineering)		
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	NASA STD 6001 Test 1 is the major method used to evaluate flammability of materials intended for use in habitable environments of U.S. spacecraft. The method is an upward flame propagation test initiated in a stagnant environment and using a well-defined igniter flame at the bottom of a vertically mounted sample. A material passes this test if the vertical burn length is less than 15.2 cm (6 inches) and there is no evidence of transfer of burning debris. The test is conducted in the most severe flaming combustion environment (oxygen concentration, pressure) expected in the spacecraft, currently 30% oxygen at 10.2 psia. Recent research has shown that current normal gravity materials flammability tests do not correlate with flammability in ventilated, micro- or reduced-gravity conditions. The materials selection for spacecraft is based on the assumption of commonality between ground test flammability results and spacecraft environments, which does not appear to be valid.		

Task Description:	To better understand the actual normal gravity flammability limits, a modified Test 1 protocol has been proposed that, as an alternative to qualifying materials as pass/fail in the worst-expected environments, measures the actual upward flammability limit (akin to the Limiting Oxygen Index (LOI)) for the material so that a more accurate assessment of the margin of safety of the material in the real use environment can be made. The method allows the option of selecting better or best materials based on the margin of safety, as opposed to what would be considered just "passing" from a flammability point of view. A research program is underway to correlate normal gravity flammability test data with data under ventilated reduced gravity or microgravity conditions. The program will determine materials suitability for microgravity and reduced gravity testing by evaluating oxygen concentration thresholds under various total pressures in normal gravity tests and in spacecraft-specific environments (ventilated microgravity or reduced partial gravity). The normal gravity data will be correlated with flammability data in microgravity environments and the safety factors involved when selecting spacecraft materials based on normal gravity testing will be evaluated. Rigorous correlations between ground test flammability data and data in ventilated spacecraft environments would allow selection of materials with increased fire resistance, and thus decrease the fire risk in space systems. The operational flexibility will also increase since the safety factors will be assessable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of the fundamental limit in microgravity and its comparison with the normal gravity flammability limit will help us understand flammability in different configurations on Earth.
Task Progress:	Currently, NASA STD 6001 Test 1 is the major method used to evaluate flammability of materials intended for use in habitable environments of U.S. spacecraft. The method is a 6” propagation pass/fail upward flame propagation test conducted over a 5 cm wide by 30 tall slabs of material exposed to the most hazardous environment (oxygen concentration, pressure) expected in the spacecraft. Flow is buoyant only (no forced flow). It is well known that in normal gravity, upward and downward flammability limits are different. The velocity and velocity gradient across surfaces is affected by the geometry of the surface. A slab has a leading edge that affects the development of the boundary layer, and the back edge step induces a recirculation zone where an opposed flow flame would stabilize; a sphere has a stagnation zone and a wake region; and a rod has a stagnation region, a boundary layer on the sides, and a wake region. Depending on where the flame ignites, the material may or may not be flammable. It is not clear, however, if this test configuration is the worst case in zero gravity, where flow velocity is decoupled from oxygen concentration, pressure, and flame temperature. To compare the effect of geometry on the flammability of materials, flat slabs, spheres, and rods have been tested in both 1g and 0g to evaluate changes in the flammability limits. Testing and analysis of the results is still underway. The goal of the geometry variation is to define the ‘worst case’ geometry for zero gravity as well as to evaluate how sensitive the flammability limits are in normal g to geometry changes. This year, the research results were presented at the Central States Section Meeting of the Combustion Institute in Champaign, IL in March, 2010, at the International Conference on Environmental Systems in Barcelona, Spain, and as a Work In Progress Poster at the International Combustion Symposium in Beijing, China in August, 2010. The Principal Investigator attended the June PI Kickoff meeting in Cleveland, OH, and the Co-Investigators visited NASA GRC in August to tour facilities and discuss test plans. The PI visited White Sands Test Facility in September, 2010 to meet with Co-Investigators to tour their facilities and discuss the research. The concept for a flight experiment is being developed.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Papers from Meeting Proceedings	Ferkul PV, Olson SL. "Zero-Gravity Centrifuge Used for the Evaluation of Material Flammability in Lunar-Gravity." 40th International Conference on Environmental Systems, Barcelona, Spain, July 11-15, 2010. 40th International Conference on Environmental Systems, July 11-15, 2010. , Jul-2010
Papers from Meeting Proceedings	Olson SL, Hirsch DB, Ferkul PV. "Improving Materials Flammability Screening for Spacecraft Fire Safety." 2010 Technical Meeting of the Central States Section of The Combustion Institute, Champaign, IL, March 22-23, 2010. Proceedings of the 2010 Technical Meeting of the Central States Section of The Combustion Institute, Champaign, IL, March 22-23, 2010. , Mar-2010